

JUANDA SURABAYA, Indonesia

WMO: 969350

Lat: 7.3747S

Lon: 112.7820E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	70.2	71.7	61.8	82.6	85.6	63.7	88.2	84.7	19.5	85.9	18.4	85.4	0.9	270	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	15.0	93.4	76.5	92.4	76.7	91.5	77.0	80.8	88.2	80.3	87.5	79.9	87.1	10.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	150.2	84.4	78.3	147.1	84.0	78.0	145.3	83.8	44.4	88.8	43.9	88.2	43.5	87.5	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.0	13.8	DB	68.2	95.2	1.9	1.3	66.9	96.2	65.8	96.9	64.8	97.6	63.4	98.5	(4)
(5)			WB	64.8	82.8	2.9	1.6	62.7	83.9	61.1	84.8	59.5	85.7	57.4	86.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	82.5	82.4	82.8	83.3	83.1	82.0	80.8	80.6	82.0	84.2	84.4	83.3
	DBStd	1.98	1.63	1.42	1.40	1.43	1.61	1.70	1.73	1.79	1.68	1.48	1.81	1.79
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11901	1008	906	1016	999	1026	959	956	949	960	1059	1032	1031
	CDD65	6426	543	486	551	549	561	509	491	484	510	594	582	566
	CDH74	71732	5313	4716	5728	6397	6747	5811	5296	5256	5920	7446	7056	6046
	CDH80	28255	1826	1579	2052	2532	2732	2193	1894	1975	2539	3469	3168	2296
Wind	WSAvg	5.6	5.8	6.1	4.7	4.8	5.4	5.7	6.1	6.7	6.6	6.2	4.8	4.8
Precipitation	PrecAvg	65.0	12.4	11.2	11.6	7.4	4.7	2.2	1.1	0.3	0.4	1.1	4.4	8.2
	PrecMax	102.7	22.5	17.9	19.5	11.7	10.2	10.0	4.9	3.5	4.4	4.8	11.6	16.4
	PrecMin	43.2	5.0	5.0	7.2	2.9	0.5	0.0	0.0	0.0	0.0	0.0	0.0	1.7
	PrecStd	16.6	4.2	3.8	3.7	3.0	3.3	2.7	1.5	0.9	1.1	1.6	3.9	3.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	91.7	91.7	91.6	91.4	90.0	89.8	90.0	92.3	94.2	95.0	93.6
		MCWB	78.9	78.7	79.1	78.6	78.4	77.2	75.9	75.3	73.5	74.7	76.5	78.8
	2%	DB	90.8	90.2	90.4	90.5	90.1	89.4	88.5	89.0	91.3	93.2	93.5	91.8
		MCWB	78.9	78.6	78.9	78.6	78.1	77.2	75.6	74.9	74.1	75.2	76.9	78.5
	5%	DB	89.5	89.2	89.4	89.7	89.5	88.4	87.9	88.1	90.1	92.1	92.4	90.6
		MCWB	78.6	78.6	78.8	78.6	78.0	76.9	75.4	74.5	74.2	75.4	77.2	78.5
	10%	DB	88.0	87.8	88.1	88.8	88.7	87.7	87.1	87.4	89.4	91.2	91.1	89.2
		MCWB	78.4	78.5	78.7	78.6	77.8	76.7	75.1	74.2	74.3	75.6	77.7	78.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	80.8	81.1	81.4	81.0	80.1	79.7	79.0	79.2	79.9	80.7	81.0
		MCDB	89.0	88.4	88.5	88.2	87.8	86.5	86.5	85.8	86.2	87.8	88.9	89.8
	2%	WB	80.0	79.9	80.2	80.6	80.3	79.5	78.7	77.9	78.1	79.1	79.9	80.1
		MCDB	87.8	87.2	87.2	87.3	87.1	85.9	85.0	85.2	85.6	87.1	88.1	88.5
	5%	WB	79.4	79.2	79.7	80.0	79.8	79.0	77.9	77.0	77.2	78.4	79.3	79.6
		MCDB	86.9	86.3	86.6	86.6	86.4	85.2	84.2	84.7	85.1	86.6	87.5	87.7
	10%	WB	78.8	78.8	79.2	79.6	79.2	78.4	77.1	76.2	76.4	77.9	78.9	79.0
		MCDB	86.1	85.8	85.8	86.0	85.6	84.6	83.5	84.1	84.7	86.2	87.1	86.7
Mean Daily Temperature Range	5% DB	MDBR	12.3	12.3	12.2	11.6	11.6	12.1	13.3	14.5	15.5	15.0	13.7	12.8
		MCDBR	13.9	13.9	13.3	12.3	12.2	12.6	13.8	14.8	16.3	16.1	15.4	14.4
		MCWBR	4.8	4.9	4.7	4.2	3.9	4.3	4.6	4.9	4.8	4.7	4.5	4.8
	5% WB	MCDBR	13.0	12.7	12.3	11.6	10.9	10.7	11.4	12.6	13.9	14.1	13.6	13.2
	MCWBR	4.9	4.9	4.9	4.4	4.0	4.1	4.3	4.5	4.6	4.5	4.7	4.9	
Clear-Sky Solar Irradiance	taub	0.534	0.535	0.527	0.509	0.478	0.453	0.440	0.458	0.475	0.544	0.583	0.568	
	taud	2.097	2.098	2.122	2.166	2.253	2.324	2.354	2.277	2.219	2.020	1.932	1.982	
	Ebn at Noon	260	259	257	253	252	254	260	263	267	254	245	250	
	Edh at Noon	54	54	52	48	42	38	37	42	47	58	63	61	
All-Sky Solar Radiation	RadAvg	1556	1588	1622	1619	1587	1554	1649	1845	2007	1943	1722	1525	
	RadStd	139	108	106	107	112	107	91	59	103	174	152	127	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.43	+1.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.43	+1.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page